Ayush Kumar

Software Engineer | Backend & Distributed Systems, Machine Learning
Computer Science & Engineering
RV Institute of Technology and Management
Location: Bengaluru, Karnataka, India (available on-site and for the full internship period)

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Portfolio Website
GitHub Profile
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Education

•RV Institute of Technology and Management *Expected 2027*
Bachelor of Engineering in Computer Science & Engineering CGPA till 5th Sem: 9.03

•Doon Public School, Dhanbad *2022*
Central Board of Secondary Education (CBSE) Percentage: 78.6%

•De Nobili School CMRI, Dhanbad *2020*
Indian Certificate of Secondary Education (ICSE) Percentage: 90.2%

Technical Skills

Languages: Python, Java, C++, C, JavaScript, TypeScript
Backend: FastAPI, Flask, Node.js, Express, REST APIs, WebSockets, Async Processing
Frontend: Next.js, React, Vite, D3.js, Tailwind CSS
Databases: PostgreSQL, MySQL (Relational), MongoDB (NoSQL)
Cloud & DevOps: Docker, AWS, Azure, Linux, Git, CI/CD Pipelines
ML & Data: PyTorch, Hugging Face Transformers, scikit-learn, NumPy, Pandas
Concepts: REST APIs, Distributed Systems, WebSockets, Event-Driven Architecture

Projects — Real-Time Systems & ML Platforms

•SENTINEL — Multi-Agent Market Microstructure Simulator with Predictive Analytics *Ongoing*
Advanced market simulation platform for liquidity crisis prediction and large order detection.

- Simulating high-frequency order book activity with **7+ trading agents**, processing thousands of events/sec with **<100ms prediction latency**.
- Engineered a Liquidity Shock Predictor providing **60–90s** early warnings before market breakdowns with **85%** accuracy validated on the **2010 Flash Crash** and **2015 ETF crash** events.
- Implemented ML-based detection of hidden **institutional orders (TWAP/VWAP/Iceberg)** achieving 83% accuracy.
- Developed and deployed a **Next.js** + **TypeScript** real-time dashboard with **WebSocket** streaming for live order book visualization and prediction monitoring.
- Tools: **Python**, **FastAPI**, **XGBoost**, scikit-learn, Next.js, TypeScript, WebSockets, LOBSTER data, NumPy, Plotly

•EQUITYFLOW — Real-Time Paper Trading for Stocks, F&O, and Commodities *Feb 2026-Mar 2026*
Full-stack trading simulator with live pricing, portfolio P&L tracking, and contract-aware F&O execution.

- Designed a real-time trading simulation engine handling live market feeds, portfolio accounting, and derivatives contract resolution.
- Implemented **live market updates** via streaming/polling pipelines, including row-level TLP refresh and directional flash highlights (green/red) on price ticks.
- Engineered portfolio/accounting logic for holdings, order history, **realized/unrealized P&L**, and average price.
- Built F&O symbol resolution layer using indexed instruments data (∼**40k+ FNO keys**).
- Tools: **Next.js**, React, **TypeScript**, **FastAPI**, **Python**, Tailwind CSS, Zustand, Recharts/Lightweight Charts, REST APIs, PowerShell scripts

•GridPulse — Formula 1 Telemetry Analytics Platform *Dec 2025*
Telemetry analytics platform for ingesting and analyzing high-frequency time-series data streams.

- Architected **FastAPI** ingestion backend service handling **5000+ datapoints/min** and stored telemetry streams in PostgreSQL for real-time analytics.
- Designed caching and streaming architecture enabling **sub-100ms** dashboard updates.
- Engineered telemetry replay pipelines synchronizing driver data streams to reconstruct race events for real-time visualization.
- Developed interactive analytics dashboards for real-time performance monitoring.
- Tools: FastAPI, React, D3.js, Docker, Data Visualization

•**Text-Based Parkinson's Disease Screening using Transformers** *Jul 2025*

Multimodal ML pipeline for clinical classification using transformers and ensemble models.

- Developed an **NLP-based diagnostic pipeline** using transformer models on 42k patient records, deploying an inference API with Flask achieving **92% accuracy**.
- Fine-tuned PubMedBERT, BioGPT, Clinical-T5 on **42K**+ patient records.
- Built reproducible training + evaluation pipelines with cross-validation and ablation studies.
- Implemented a production-ready inference API using Flask for clinical prediction services.
- Tools: PyTorch, Transformers, LightGBM, Flask, CUDA

•**Occasio — Event Management & Booking Platform** *Oct 2025–Nov 2025*

Event ops platform for organizers with event publishing, registrations, payments, ticketing, check-in, analytics, team access, and certificates.

- Built a React + Vite frontend for public/admin flows, including event discovery, registration, and scanner check-in.
- Implemented Node.js + Express + Prisma backend for event CRUD, auth, analytics, broadcast messaging, team roles, and ticket/wallet APIs.
- Added organizer features like ticket style builder, certificate designer (multi-type templates + preview + distribution), and reminders/polls/reviews.
- Integrated payment and fulfillment with queue-based ticket/email processing and downloadable tickets.
- Prepared for production with PWA support, env-based API config, and cloud-ready frontend/backend separation.
- Tools: React, Vite, Tailwind CSS, Node.js, Express, Prisma, PostgreSQL, BullMQ/Redis, PDF + QR workflows

Positions of Responsibility

•**Technical Head** Robotics, Web Dev, Cybersecurity, Coding and Entrepreneurship club *2025–Present*

•**Events and Website Head** Core Organizing Committee, National Level Techfest Robofiesta 2025 *2025–Present*

•**Core Team Member** Robotics, Web Dev, Cybersecurity, Coding, GDG and Entrepreneurship club *2025–Present*

Achievements

•**Codeforces Contest** Ranked top 20% in global Codeforces contest (52k+ participants) worldwide. *Dec 2024*

•**Competitive Programming** Solved 300+ algorithmic problems across Codeforces, LeetCode, CodeChef, and AtCoder. *Ongoing*

Open Source Contributions

•**CDLI (Cuneiform Digital Library Initiative)** Contributed 6+ merge requests on GitLab improving repository tooling and data processing workflows. *Mar 2026*

•**Gemini CLI (Google)** Contributed 4+ pull requests improving CLI tooling, including WSL shell execution routing fixes and trusted tool configuration enhancements. *Mar 2026*

•**Processing Foundation** Contributed pull request fixing UI bug in Processing website repository, resolving incorrect search placeholder behavior. *Dec 2023*

Research & Publications

•**Kumar, A.**, et al. (2025). *Optimizing Soybean Production: A Data-Driven Machine Learning Framework for Disease Prediction and Yield Improvement.* Zenodo (preprint). https://doi.org/10.5281/zenodo.15788313

•**Kumar, A.**, et al. (2025). *Context is All You Need: Why Transformers Outperform Traditional ML Models in Tweet Sentiment Classification.* Zenodo (preprint). https://doi.org/10.5281/zenodo.17606042